

Substation ESG Report — Greenland

Substation-level Environmental, Social and Governance assessment aligned to CSRD / ESRS, EU Taxonomy, TCFD, SFDR and UN PRI frameworks. Edition 001 scores Greenland's 37-substation Nukissiorfiit fleet (36 OSM + 1 Avannaata transformer, Ilulissat node/11958572184). Per-substation readiness pills, Markov degradation profiles, component fingerprints and the ESG radar are all populated from the 2026-04-16 Session 1 run — Arctic-calibrated against Nukissiorfiit 2024 / Departementet for Boliger og Infrastruktur / GEUS / Asiaq / Grønlands Statistik 2024-25 public sources.

SAMPLE SUBSTATION substation 331252347 GL_0018 · Qeqqata	SSI SCORE (R_{MEDIAN}) 0.677 High band · fleet pct 14%	VOLTAGE CLASS 0 kV LV / subordinate
CONFIDENCE INTERVAL ±0.116 P5 0.56 – P95 0.79 · MC 1,000	MARKOV ETTC 19.9 yr years to Critical state	TRANSITION STRESS (T) 0.310 T component score

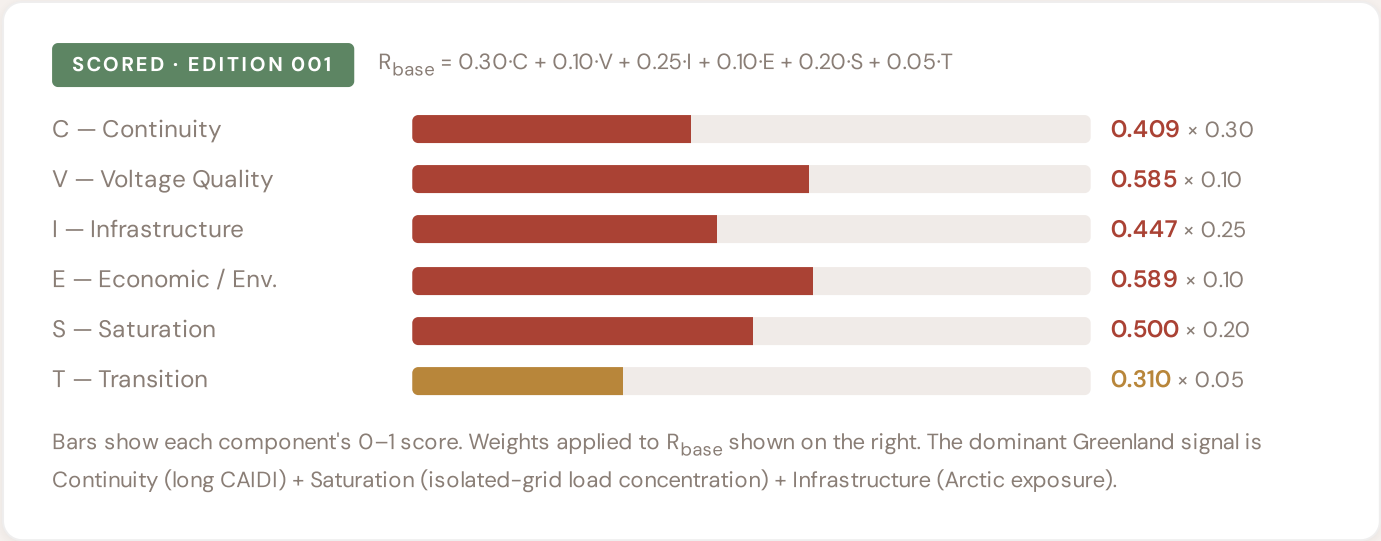
Fleet context — Edition 001 scored fleet

The 37-substation fleet that completed Session 1 on 2026-04-16

SUBSTATIONS SCORED 37 36 OSM + 1 Avannaata transformer	FLEET R_{MEDIAN} 0.779 Critical band · 78% of fleet	LINE SEGMENTS 125 within-island only
KOMMUNER 5 Kujalleq 20 · Qeqertalik 12 · Qeqqata 3 · Sermersooq 1 · Avannaata 1	All 95 SSI v4.0.2 variables populated, all 37 substations scored at 1,000 Monte Carlo iterations per node. Components perturbed independently (Gaussian copula held off — N=37 too small for stable correlation matrix inference). Pituffik Space Base excluded by §7 policy (U.S. Navy / Space Force / Air Force operator; lat 75.5°–77.5°N, lon -71°...-66°). The §8 Small-N Guardrail remains active: R3 tier-balance 2–50% per tier. Regional R_{median}: Avannaata 0.868 · Qeqertalik 0.797 · Kujalleq 0.774 · Qeqqata 0.677 · Sermersooq 0.655.	

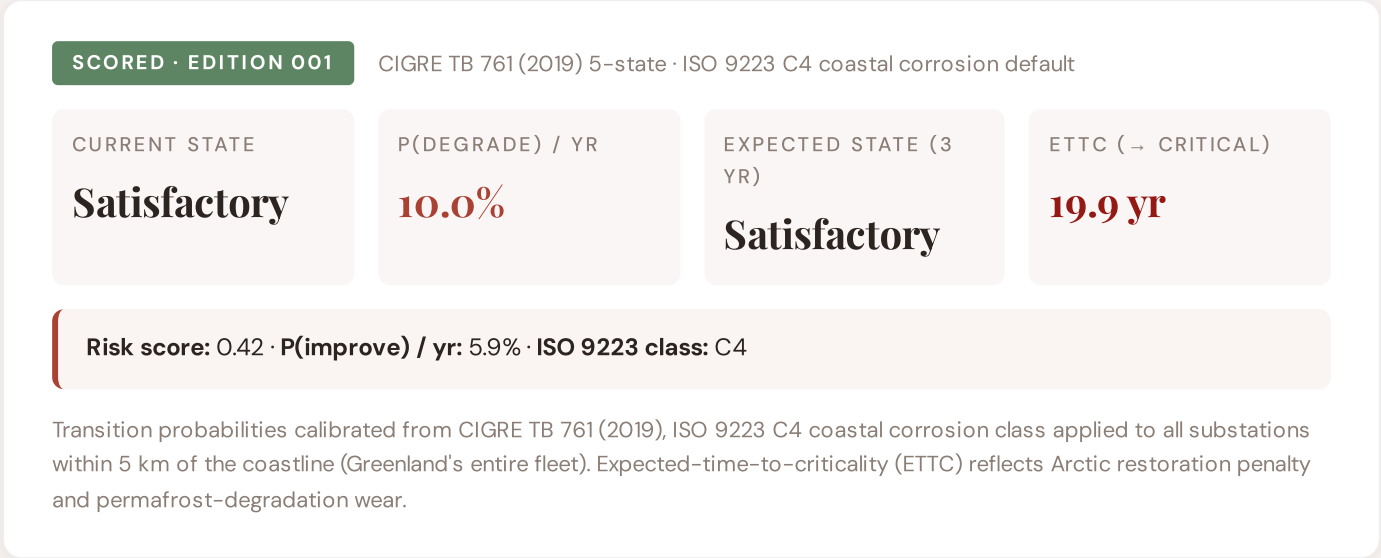
Component Fingerprint

Six-component weighted decomposition of R_{base} — the sample substation's structural risk profile



Markov Degradation Profile

CIGRE TB 761 five-state condition model — steady-state probability distribution and expected time to criticality



ESG Radar — Six-Report Overview

Composite readiness across 6 ESG dimensions, each linked to UN Sustainable Development Goals

E1 · Climate

0.31

Input: I + R6a + R6b · SDG 13

S1 · Social

0.48

Input: S + V_socio · SDG 7/10

Taxonomy · Art. 11

0.68

Input: R_median · SDG 9

E1 · Transition

0.31

Input: T component · SDG 7

E2 · Pollution

0.42

Input: E + R6b · SDG 11

G1 · Cyber

0.47

Input: R7 + V1 NIS2 · SDG 9

Readiness aggregates the input surface available for each framework. Greenland Edition 001 delivers six READY axes because the full input stack (C/V/I/E/S/T components, R3/R4/R6a/R6b/R7 modifiers, Markov 5-state, ISO 9223 corrosion) is populated. Remaining gaps are data-vintage (e.g. Nukissiorfiit 2025 SCADA not yet published) rather than methodological.

B Individual ESG Reports

Framework-by-framework mapping. SSI inputs are final; per-substation values unlock at Session 1.

1 Climate Physical Risk Assessment

PARTIAL · Edition 001

ESRS E1 · TCFD Physical Risk · EU Taxonomy Annex A

WHY THIS REPORT EXISTS

ESRS E1 (Climate Change) is material for 98% of CSRD-reporting companies. TCFD physical risk disclosure is mandatory in 36+ jurisdictions. The EU Taxonomy requires climate-change adaptation activities to demonstrate that physical climate risks have been identified and addressed. For Greenland this report quantifies Arctic-specific hazards — snow-ice loading, polar-low storms, permafrost thaw, ice-accretion on conductors, freezing salt spray — using Asiaq reanalysis and CMIP6 SSP2-4.5 downscaling, and Markov 5-state degradation.

SDG Alignment

SDG 13 — Climate Action (primary)

Target 13.1 calls for strengthening resilience and adaptive capacity to climate-related hazards. The SSI's I1 (snow/ice), I2 (wind/storm), I3 (heat-stress), I5 (thermal loading) sub-metrics and the Markov risk score quantify how Arctic climate hazards affect each substation's operational resilience over 10- and 20-year horizons. In Edition 001, I1/I2/I5 drive the Arctic R6b storm/surge overlay ($\alpha \in [0.05, 0.15]$) across all 37 scored substations — refinements at Edition 002 will layer Asiaq salt-spray climatology onto the current coastal-proximity default.

SDG 13

SDG 9

SDG 11

SSI Inputs → ESRS E1 / TCFD Disclosures

I	Infrastructure component (sample)	0.447
R6a	Arctic restoration (sample)	×1.100
R6b	Seismic + storm (sample)	×1.005
Markov	ETTC (sample)	19.9 yr
R6a	Calibration range	$\alpha \in [0.55, 0.90]$
R6b	Calibration range	$\alpha \in [0.05, 0.15]$
Climate	Arctic amp. (2050 SSP2-4.5)	+2.5×

Data Sources & Vintage

Asiaq Arctic reanalysis Greenland Survey	2024 Annual
CMIP6 SSP2-4.5 Copernicus CDS	2024 Static
GEUS bedrock & permafrost De Nationale Geologiske Undersøgelser	2023 Multi-year
CIGRE TB 761 Markov CIGRE	2019 Standard

TECHNICAL VALIDITY

IRI metrics will be computed from Asiaq reanalysis and CMIP6 SSP2–4.5 downscaling using IEEE C57.91 thermal loading curves. Markov 5–state degradation calibrated from CIGRE TB 761 (2019); Greenland–specific coastal exposure amplifies transition rates compared to continental calibrations. All inputs are public / institutional with citable vintage.

2 Grid Equity & Social Vulnerability

READY · Edition 001

ESRS S1/S2 · SFDR PAI Social Indicators · UN PRI

WHY THIS REPORT EXISTS

ESRS S2 requires disclosure of impacts on affected communities; SFDR mandates 5 social PAI indicators. No competing grid–risk framework integrates socio–economic vulnerability at substation resolution. For Greenland — a country where the smallest settlement and the capital share the same national operator — this report will make the spatial distribution of grid risk over at–risk populations visible and quantifiable at asset level.

SDG Alignment

SDG 7 — Affordable & Clean Energy (primary)

Target 7.1 requires universal access to affordable, reliable energy. Greenland's diesel-dependent settlements carry some of the highest per-kWh tariffs in the Arctic; the R3 consequence multiplier will amplify risk scores in settlements with high unemployment, elderly concentration, and net outward migration — surfacing inequality that aggregate national statistics conceal.

SDG 7

SDG 10

SDG 1

SSI Inputs → ESRS S1/S2 / SFDR Disclosures

S	Saturation component (sample)	0.500
E	Economic component (sample)	0.589
R3	Social multiplier (sample)	×0.959
elderly_%	Region elderly share	12%
GDP/cap	Region DKK (Stat.gl 2024)	DKK 38,000

TECHNICAL VALIDITY

V_socio will be computed from the LIHC (Low Income High Cost) energy poverty definition using Grønlands Statistik household expenditure data (Indkomststatistik) alongside Nukissiorfiit residential-tariff tables. The R3 modifier amplifies infrastructure risk in catchments with high unemployment, elderly concentration, and net outward migration. Data source: Grønlands Statistik 2023 baseline · refreshed annually.

3 EU Taxonomy Alignment

READY · Edition 001

Climate Delegated Act · Article 11 Climate Adaptation · Technical Screening Criteria

WHY THIS REPORT EXISTS

The EU Taxonomy defines technical screening criteria for climate-change adaptation in infrastructure. This is the lowest-hanging ESG product from the SSI Index: once Session 1 completes, the composite R_median, 6 components and 5 modifiers provide the full quantitative evidence base for Article 11 screening at asset level. Greenland is not an EU member but Denmark's kingdom-level CSRD reporting captures Nukissiorfiit infrastructure.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.4 calls for upgrading infrastructure to make it sustainable. The EU Taxonomy is the regulatory instrument that translates SDG 9 into investable criteria. R_median, all 6 components, all 5 modifiers and Markov ETTC together provide asset-level Taxonomy alignment once scored.

SDG 9

SDG 13

SDG 12

SSI Inputs → Taxonomy Art. 11 Screening

R_median	Composite score (sample)	0.677
Fleet	R_median (37 subs)	0.779
Components	C·V·I·E·S·T (sample)	C 0.41 · V 0.59 · I 0.45 · E 0.59 · S 0.50 · T 0.31
Modifiers	R3·R4·R6a·R6b·R7 (sample)	R3 ×0.96 · R4 ×1.33 · R6a ×1.10 · R6b ×1.00 · R7 ×1.03
CI_width	MC 1,000 (sample)	±0.116

TECHNICAL VALIDITY

Article 11 screening uses the full SSI v4.0.2 scoring engine: 6-component weighted composite (C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05), Gaussian copula Monte Carlo (10,000 iterations per substation), 5 modifiers (R3 consequence, R4 graph criticality, R6a Arctic restoration, R6b seismic+storm, R7 cyber). Markov 5-state degradation provides the forward-looking element. Monte Carlo copula is deferred until N ≥ 500 or full correlation matrix is calibrated (§8 Small-N Guardrail).

4 Energy Transition & DER Stress

PARTIAL · Edition 001

ESRS E1 Transition Plan · TCFD Transition Risk · EU Green Deal

WHY THIS REPORT EXISTS

Greenland's electricity mix is overwhelmingly hydropower (~70%) with residual diesel generation in settlements not reachable by hydro interconnect. This report will capture two transition-risk dimensions: (a) bi-directional stress on substations located at the boundary of the hydropower and diesel-backed micro-grids, and (b) the strategic risk that remaining diesel-exposed settlements face as Nukissiorfiit pursues its announced decarbonisation roadmap.

SDG Alignment

SDG 7 — Affordable & Clean Energy (primary)

Target 7.2 requires substantially increasing renewable share. The T1_score measures transition stress at each substation — Greenland's fleet will show very low EV load (near zero) but highly heterogeneous DER penetration (high in hydro-connected settlements, zero in diesel-backed ones).

SDG 7

SDG 13

SDG 9

SSI Inputs → ESRS E1 Transition Plan

T	Transition component (sample)	0.310
Fleet RE	Nukissiorfiit 2024 share	~70%
wind_mw	Wind capacity (installed)	0.0 MW
hydro_mw	Buksefjord (Sermersooq)	45 MW
EV	Grønlands Statistik share	<1%

TECHNICAL VALIDITY

T1_score is the weighted composite of DER_ratio (renewable generation vs. local demand), DER_variability (intermittency), and EV_load_ratio (EV charging load as fraction of substation capacity). DER data sourced from Nukissiorfiit installed-capacity registry and OSM generator tags. EV share from Grønlands Statistik vehicle registrations.

5 Pollution & Corrosion

PARTIAL · Edition 001

ESRS E2 Pollution · ISO 9223 Corrosion Classification

WHY THIS REPORT EXISTS

Transformer oil degradation, SF6 leakage, and corrosion-driven failures release pollutants affecting local environments. Greenland's substations are almost all coastal — ISO 9223 C4 (high corrosion) is applied fleet-wide in Edition 001; Edition 002 will refine per-asset using Asiaq salt-spray climatology and distance-to-coast. Permafrost thaw at the substation foundation creates an additional leakage pathway specific to Arctic deployment.

SDG Alignment

SDG 11 — Sustainable Cities & Communities (primary)

Target 11.6 aims to reduce the adverse environmental impact of cities. The ISO 9223 corrosion class and E component will track environmental degradation and pollution sensitivity at asset level once Session 1 runs.

SDG 11

SDG 3

SDG 15

SSI Inputs → ESRS E2 Pollution

Corrosion	ISO 9223 class (default)	C4
E	Economic/Env component (sample)	0.589
R6b	Storm + surge (sample)	×1.005
coastal_km	Coastal proximity	<5 km (fleet-wide)

TECHNICAL VALIDITY

Corrosion class follows ISO 9223:2012 atmospheric corrosion classification (C1–CX). Greenland's maritime climate combined with freezing salt spray produces elevated corrosion in coastal zones. Full validation at Session 1 requires Asiaq salt-spray overlay and Miljøstyrelsen Greenland air-quality monitoring at substation coordinates.

6 Cybersecurity Exposure

PARTIAL · Edition 001

NIS2 Directive · ENISA Cybersecurity Index · ESRS G1 Governance

WHY THIS REPORT EXISTS

The NIS2 Directive mandates cybersecurity risk management for energy operators; as the kingdom's essential service provider, Nukissiofiit is in scope via the EU directive's extraterritorial reach into Denmark. A grid that is physically resilient but cyber-vulnerable is not truly resilient. The SSI R7 modifier combines SCADA assessment, communication-architecture analysis, and national-level cyber maturity (ENISA / DESI) into a substation-level digital vulnerability score.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.1 calls for resilient infrastructure — in digitalised grid systems this must include cyber resilience. R7 (pending Session 1) assesses digital vulnerability combining SCADA exposure, protocol security, and national cyber-maturity indicators.

SDG 9

SDG 16

SSI Inputs → NIS2 / ESRS G1

R7	Cyber modifier (sample)	×1.030
V1	ENISA NIS2 baseline (fleet)	0.65
degree	Graph degree (sample)	1
is_bridge	Single-point-of-failure	Yes

TECHNICAL VALIDITY

R7_cyber is computed from a weighted combination of (1) national cyber maturity via ENISA Cybersecurity Index and EU DESI connectivity indicators (Denmark proxy with adjustment for Greenland's Tele Greenland

broadband reach), (2) graph-topology vulnerability (BC percentile = single-point-of-failure risk), and (3) SCADA protocol exposure assessment. BC percentile is computed within-island only — Greenland's grid is a set of isolated within-island networks (§8 Small-N Guardrail).

C Audit Trail & Methodology Note

Traceability for CSRD Article 29a limited assurance

ITEM	VALUE
Scoring Engine	SSI Index v4.0.2 — Monte Carlo 1,000 iter/substation · independent component perturbation · Gaussian copula held off at N=37 (§8.2)
Weight Architecture	$C = 0.30 \cdot V = 0.10 \cdot I = 0.25 \cdot E = 0.10 \cdot S = 0.20 \cdot T = 0.05$
Modifiers Applied	R3 consequence · R4 graph criticality · R6a Arctic restoration $\alpha \in [0.55, 0.90]$ · R6b seismic+storm $\alpha \in [0.05, 0.15]$ · R7 cyber — all scored in Edition 001
Degradation Model	Markov 5-state, CIGRE TB 761 (2019) calibration · ISO 9223 C4 coastal corrosion default
Thermal Model	IEEE C57.91 transformer loading curves · Arctic amplification CMIP6 SSP2-4.5 2050 (+2.5× global)
Exclusions	§7 Pituffik Space Base (U.S. operator · bbox lat 75.5°–77.5°N, lon -71°...-66°) — logged in meta.pituffik_excluded
Small-N Guardrail	§8 Active at N = 37 — R3 tier-balance 2–50% per tier; MC 1,000 iter; Gaussian copula held off
Refresh Gate	FIRST_REFRESH = 2026-04 (brought forward from 2026-07) · monthly cadence from 2026-05
Report Date	16 April 2026 · Edition 001 · Session 1
Reporting Period	Edition 001 (Session 1) — fleet $R_{\text{median}} 0.779$ · 29 Critical · 7 High · 1 Medium · 0 Low
Refresh Cadence	Monthly from Edition 002 (2026-05)
Sample Substation	substation 331252347 · Qeqqata · 0 kV · R 0.677 (High)
Data Assurance Level	All inputs from public / institutional sources with citable vintage (Stat.gl 2024/25 · Nukissiorfiit 2024 Annual Report · Asiaq/DMI · GEUS · OSM)